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Emergent Narrative using Incremental Story Sifting

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- A clear statement of the problem and goals of the project
- A review of the background literature
- An analysis and specification of the solution to the problem
- A detailed design
- The implementation
- Testing strategy and results
- A critical evaluation
- Conclusions and future work
- References to the literature
- Appendices providing detailed technical references

The final report should document the project planning. In particular it should compare the project plans set out in the project brief and progress report with the actual achievements. To this end it should include the following:

- A copy of the original project brief as an appendix
- One or more Gantt's chart (or PERT's chart) showing the planned schedule and the actual progress

1 Introduction

Interactive Digital Narrative (IDN) is a widely used concept, covering mediums from literacy work, artistic representations to computer science. [3] It concerns the portrayal of complex narratives, with the goal of enabling them to become more immersive for the person experiencing them. [3] [8] These techniques make the audience feel more like a participant, while the author takes less control.

IDN can be used for storytelling, for example in Netflix's *Bandersnatch*, the viewer can decide on actions which then determine which video fragments play; and in hypertext literature such as Michael Joyce's *Afternoon: A Story*, readers can navigate branching narratives in the prose via clicking embedded hyperlinks. There are three different main approaches to this type of storytelling: designed, generative, and emergent.

A *designed narrative* generally would include a structured story line, and may involve choices and alternative structured routes within. The story may be made up of connected nodes which the player can traverse in order to decide which story they will follow. [3] (*Bandersnatch* and *Afternoon: A Story* being examples of this).

Generative narrative typically does not involve a pre-structured story. Rather there are story sections, also known as story beats, which can be chosen by the system in response to earlier actions completed by the player in order to build up a personalised story. This method begins to provide more agency to the player. [2] An example of this method is *Facade*; A drama manager is used to sequence the story beats in the pool so that a story can be generated following an arc for rising and falling tension. [1]

Lastly, *emergent narrative (EM)* is an approach offering the most agency to the player, which I will be looking at further in this report. EM is a method of storytelling which arises bottom up through computer simulations about character activity. [4] [6] The story emerges entirely through play, or simulation, as events occur. The issue is these events can be seemingly random and together may not be of any *narrative interest*. Narrative interest can consist of concepts such as event relatedness or dramatic impact. People typically find stories more interesting where events share the same characters; may cause other events to occur, or events share the same goal or intention. [5]

This is where *story sifting* comes in. [4] Story sifting is a technique whereby a system can find all occurrences of story patterns that emerge within the events of the simulation. These patterns are defined in blocks called sifting patterns that can be queried over the logs generated by the simulation to return all instances that they occurred. The majority of story sifting techniques currently that have been explored are retrospective [7]. The patterns are matched and ranked after the simulation has completed and potential stories that occurred are found and reported on at the end.

1.0.1 Problem

However, there are a few limitations to story sifters. There can be little *agency* with the retrospective method as the player does not have a strong effect on the narrative within the game through their actions. Story sifting can extract the patterns, however it does nothing in the way of generating what will occur and therefore would not make for good game play alone. Furthermore, story sifters alone are unable to determine what a human will find interesting. [9]

Most of the pleasure that comes from stories is due to the anticipation of events that occur. [7] Therefore these issues would need to be addressed to add more narrative interest to system to ensure there is enough authoring implemented into the game for the appearance of a story, while also increasing agency of the player so they can affect the story - a concept known as the Narrative Paradox.

1.0.2 Goals

A newer technique that has been less looked into is incremental prospective story sifting [10]. Incremental sifting is about running the sifting engine during the course of the simulation to find the patterns beginning to emerge before they have completed. [7] This method of incremental sifting can 'see' the stories that could be occurring and, combined with an intervention method, could ensure

the most interesting of them get completed.

I am looking to design an emergent narrative game, using incremental prospective story sifting to search for patterns in a simulation the player is currently experiencing, determine which would make a good story, and therefore what further actions would be needed for it to be completed in a satisfying way.

I propose combining this solution with an influencing system to persuade the player to take certain further actions to complete these stories, and allow them to anticipate future events and endings. This should make for a game with more agency while still providing an interesting story.

To evaluate this system, I will look into testing two questions:

1. Does my system give more completed stories when the game is played compared to when my system is not in use and the game is played?
2. And does the player perceive stories occurring and do they notice they are being influenced into finishing them?

The first question will be a quantitative analysis based on how many patterns are recognised and completed when the game is run. The second question will be a qualitative analysis based on anonymous interviews with participants after they play the game.

2 Background Literature

3 Implementation

4 Evaluation

5 Conclusion

References

- [1] Michael Mateas and Andrew Stern. “Integrating Plot, Character and Natural Language Processing in the Interactive Drama Façade”. In: *Proceedings of the Technologies for Interactive Digital Storytelling and Entertainment (TIDSE) Conference*. 2003.
- [2] Mark O. Riedl. “Incorporating authorial intent into generative narrative systems”. In: *AAAI Spring Symposium - Technical Report*. Vol. SS-09-06. 2009.
- [3] Hartmut Koenitz et al. *Interactive digital narrative: History, theory and practice*. 2015. DOI: 10.4324/9781315769189.
- [4] James Ryan. “UC Santa Cruz UC Santa Cruz Electronic Theses and Dissertations Title Curating Simulated Storyworlds”. In: (2018).
- [5] Max Kreminski, Melanie Dickinson, and Noah Wardrip-Fruin. “Felt: A simple story sifter”. In: *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Vol. 11869 LNCS. 2019. DOI: 10.1007/978-3-030-33894-7{_}27.
- [6] Max Kreminski, Noah Wardrip-Fruin, and Michael Mateas. “Toward example-driven program synthesis of story sifting patterns”. In: *CEUR Workshop Proceedings*. Vol. 2862. 2020.
- [7] Max Kreminski, Melanie Dickinson, and Michael Mateas. “Winnow: A Domain-Specific Language for Incremental Story Sifting”. In: *17th AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment, AIIDE 2021*. 2021. DOI: 10.1609/aiide.v17i1.18903.
- [8] Hartmut Koenitz, Jonathan Barbara, and Mirjam Palosaari Eladhari. *Interactive digital narrative (IDN)—new ways to represent complexity and facilitate digitally empowered citizens*. 2022. DOI: 10.1080/13614568.2023.2181503.
- [9] Max Kreminski et al. “Select the Unexpected: A Statistical Heuristic for Story Sifting”. In: *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Vol. 13762 LNCS. 2022. DOI: 10.1007/978-3-031-22298-6{_}18.
- [10] Ben Clothier and David E. Millard. “Awash: Prospective Story Sifting Intervention for Emergent Narrative”. In: *Interactive Storytelling*. Ed. by Lissa Holloway-Attaway and John T. Murray. Cham: Springer Nature Switzerland, 2023, pp. 187–207. ISBN: 978-3-031-47655-6.